

CLIs or command-line *interfaces* and GUIs or *graphical user interfaces* are two of the main ways to interact with a computer. These two interface types are very different in the ways they command a system and in their abilities.

CLIs are a text-based interface. They are operated through commands which get typed as a line of text into a terminal. This creates an input output cycle, one inputs a command and the system executes accordingly. Through CLIs one can run programs, automate workflows, chain commands, access system resources, and navigate file systems. Professionally CLIs are the preferred way of interacting with a machine due to its speed and precision over GUIs. CLIs let you use chain commands and automate workflows using scripting. These interfaces use *shells*, which are command-line interpreters/processors.

GUIs let people interact with a computer using visuals. These interfaces do not require much technical knowledge and are relatively simple and intuitive to use. GUIs use cursors, buttons, icons, and menus to interact and compute commands to the operating system. GUIs work through input controls and navigation and informational components.

GUIs are used for everyday computing, anytime you complete a google search or work on an assignment in canvas we are interacting and sending commands to the system through the GUI. These interfaces are less efficient, require more computer resources, have lower rendering and precision than CLIs. However, they are incredibly helpful for day-to-day work and easy visualization for things like multitasking. CLIs are more complex and require more specialized knowledge to interact with. They are not used daily and sometimes never by most. These interfaces use less processing power, have high precision (but also require higher user precision), fast execution, and possess many command capabilities.